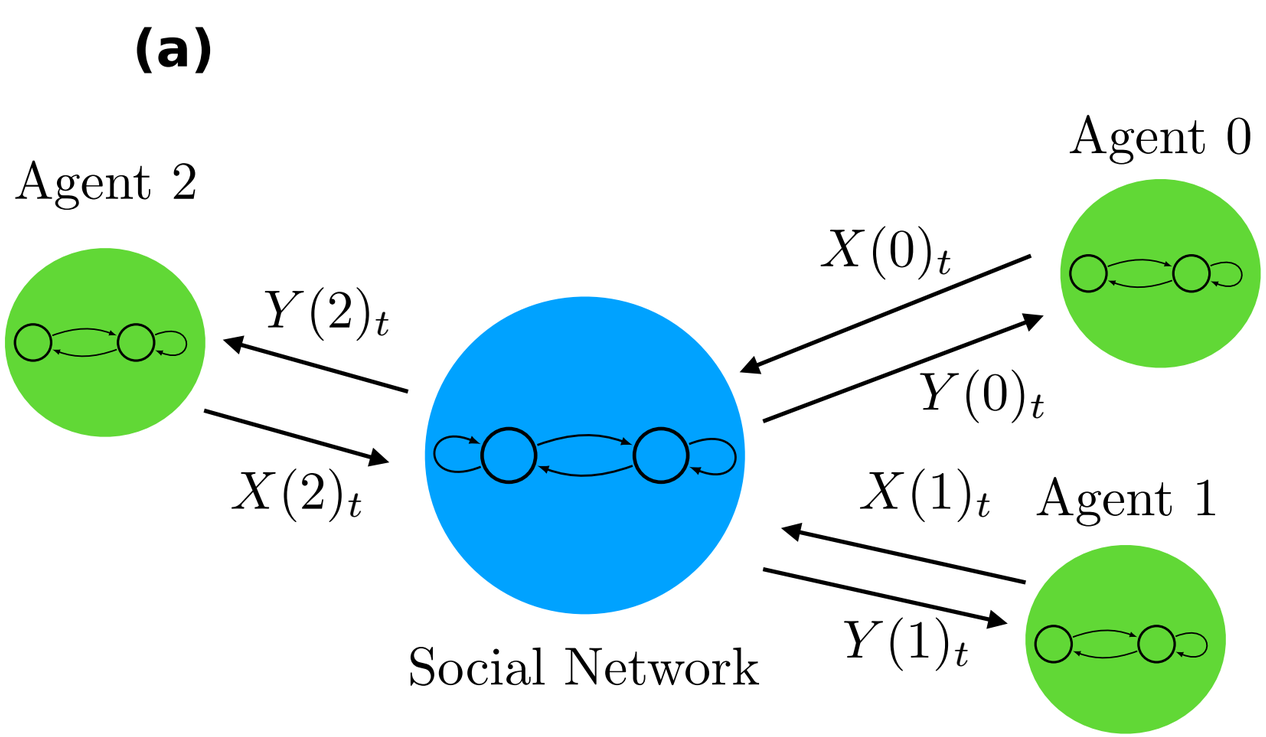
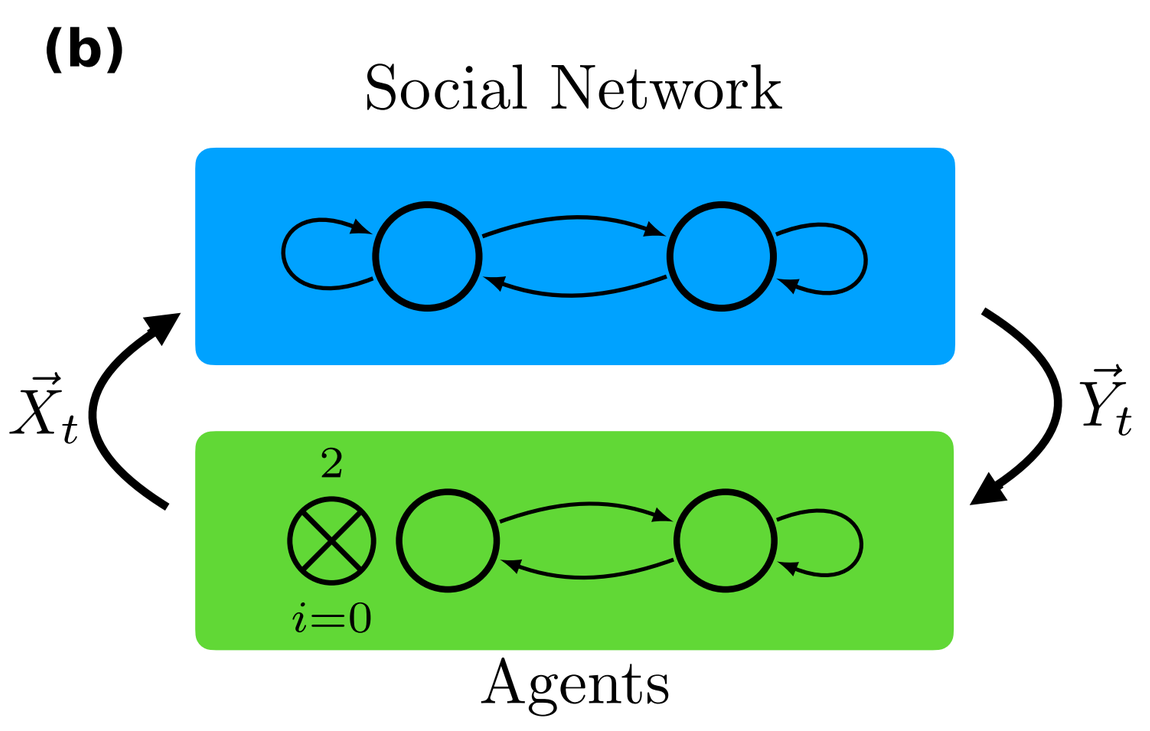
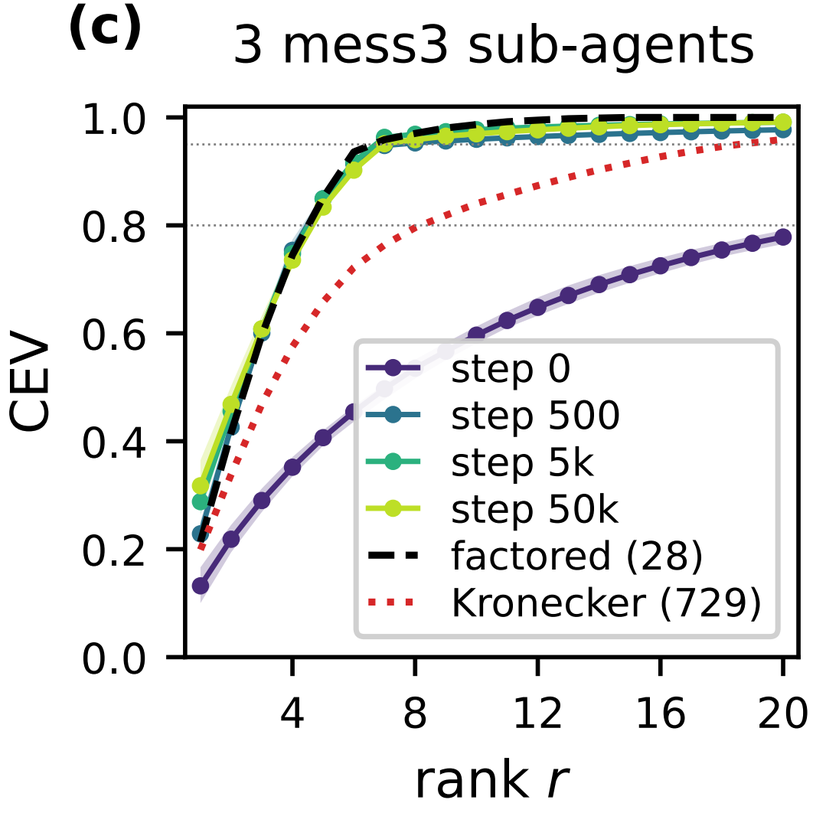
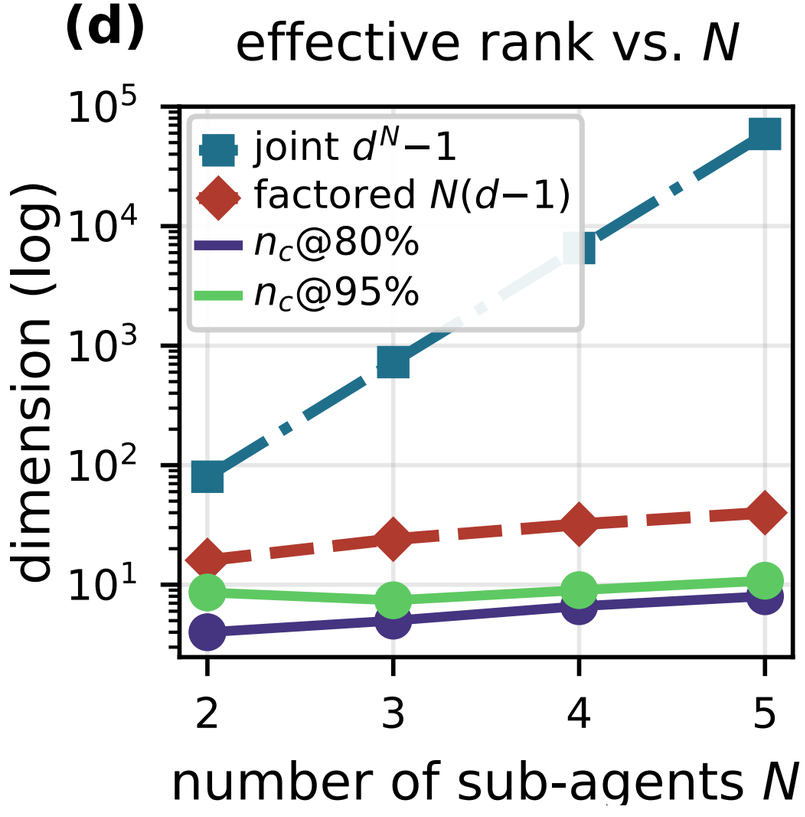
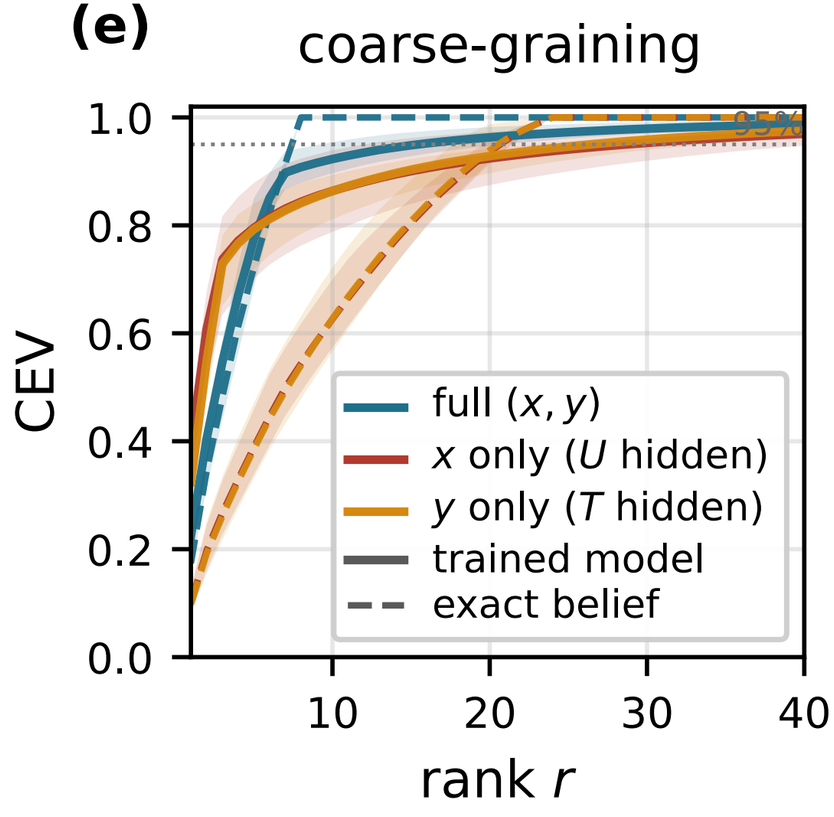

(a)
Agent 2
Agent 0
Agent 1
$Y(2)_t$
$X(2)_t$
$X(0)_t$
$Y(0)_t$
$X(1)_t$
$Y(1)_t$
Social Network


(b)
Social Network
$\vec{X}_t$
$\vec{Y}_t$
2
i=0
Agents


(c)
3 mess3 sub-agents
CEV
rank r
step 0
step 500
step 5k
step 50k
factored (28)
Kronecker (729)


(d)
effective rank vs. N
dimension (log)
number of sub-agents N
joint d^N-1
factored $N(d-1)$
n_c@80%
n_c@95%


(e)
coarse-graining
CEV
rank r
95%
full (x, y)
x only (U hidden)
y only (T hidden)
trained model
exact belief